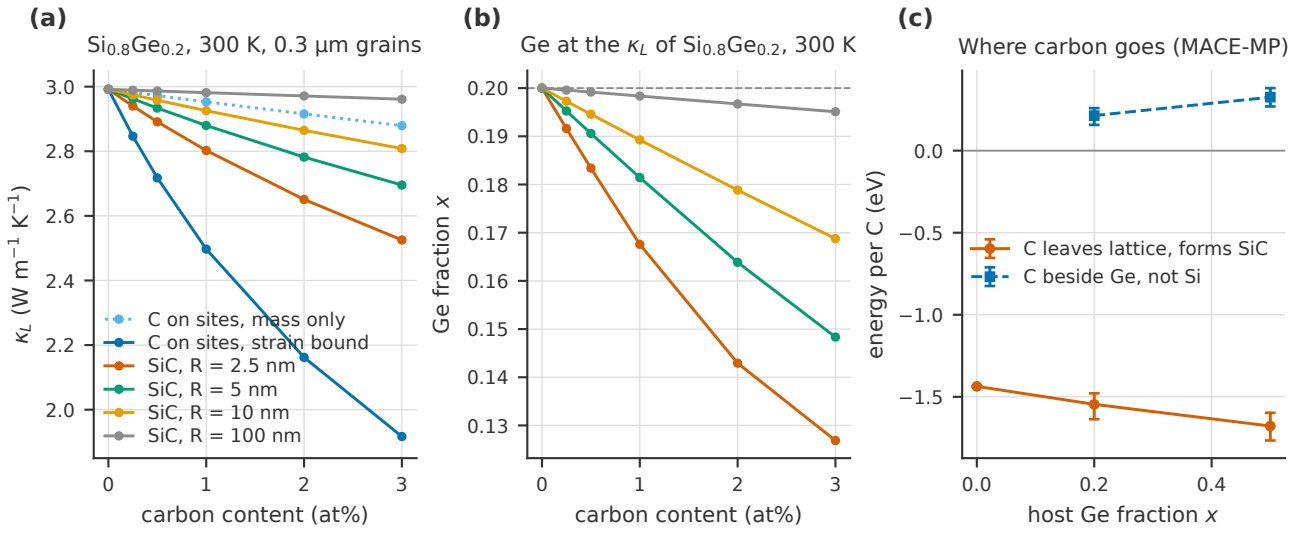
(a)
Si0.8Ge0.2, 300 K, 0.3 μm grains
κL (W m−1 K−1)
3.0
2.8
2.6
2.4
2.2
2.0
C on sites, mass only
C on sites, strain bound
SiC, R = 2.5 nm
SiC, R = 5 nm
SiC, R = 10 nm
SiC, R = 100 nm
0
1
2
3
carbon content (at%)
(b)
Ge at the κL of Si0.8Ge0.2, 300 K
Ge fraction x
0.20
0.19
0.18
0.17
0.16
0.15
0.14
0.13
0
1
2
3
carbon content (at%)
(c)
Where carbon goes (MACE-MP)
energy per C (eV)
0.0
−0.5
−1.0
−1.5
C leaves lattice, forms SiC
C beside Ge, not Si
0.0
0.2
0.4
host Ge fraction x